LLM Application Lifecycle - Complete Tools Ecosystem

Forward Deployed Engineer's Complete Arsenal - From Data to Production

PHASE 1: DATA & PREPARATION

1. Data Versioning & Dataset Management

DVC (Data Version Control)
✓ Version large datasets
✓ ML experiment reproducibility
✓ Git-like dataset tracking
Type: Open Source

LakeFS
✓ Data lake version control
✓ Enterprise dataset management
✓ Git-style branching for data
Type: Open Source + Enterprise

Hugging Face Datasets
✓ Public dataset access
✓ Dataset preprocessing
✓ Easy transformer integration
Type: Open Source

2. Data Quality & Cleaning

Cleanlab
✓ Detect mislabeled data
✓ Find noisy samples
✓ Improve dataset quality
Type: Open Source + Enterprise

Great Expectations
✓ Data validation pipelines
✓ Quality checks
✓ Automated dataset testing
Type: Open Source + Cloud

3. Additional Data Tools

Pandas
✓ Data manipulation
✓ Data analysis
Type: Open Source

Polars
✓ Fast dataframes
✓ Large datasets
Type: Open Source

DuckDB
✓ SQL analytics
✓ Embedded database
Type: Open Source

PHASE 2: DEVELOPMENT & TRAINING

4. Training & Fine-Tuning

Hugging Face Transformers
✓ Train transformer models
✓ Fine-tune LLMs
✓ Industry standard library
Type: Open Source

PyTorch
✓ Deep learning research
✓ Custom neural networks
✓ Flexible framework
Type: Open Source

DeepSpeed
✓ Large-scale model training
✓ Distributed GPU training
✓ Memory optimization
Type: Open Source

TensorFlow
✓ Production ML
✓ Enterprise deployment
Type: Open Source

5. LLM Application Frameworks

LangChain
✓ RAG pipelines
✓ LLM agents
✓ Tool integrations
Type: Open Source

LlamaIndex
✓ Connect LLMs with data
✓ Build RAG systems
✓ Document indexing
Type: Open Source

Haystack
✓ Enterprise search systems
✓ Document QA pipelines
✓ Strong retrieval features
Type: Open Source

Semantic Kernel
✓ Microsoft framework
✓ AI orchestration
Type: Open Source

6. Prompt Management & PromptOps

PromptLayer
✓ Prompt version tracking
✓ Monitoring
Type: Paid

Promptfoo
✓ Prompt testing
✓ LLM evaluation
Type: Open Source

Humanloop
✓ Prompt experimentation
✓ Human feedback
Type: Paid

PHASE 3: INFRASTRUCTURE & DEPLOYMENT

7. Vector Databases (RAG Systems)

Pinecone
✓ High-performance search
✓ Production RAG
✓ Fully managed
Type: Paid Cloud

Weaviate
✓ Hybrid search
✓ Knowledge graphs
Type: Open Source + Cloud

FAISS
✓ Fast similarity search
✓ Local vector storage
Type: Open Source

Qdrant
✓ Scalable vector search
✓ Metadata filtering
Type: Open Source + Cloud

ChromaDB
✓ Embeddings database
✓ Developer-friendly
Type: Open Source

Milvus
✓ Cloud-native vector DB
✓ High scalability
Type: Open Source

8. Deployment & Serving

FastAPI
✓ Build LLM APIs
✓ Lightweight services
✓ Fast Python framework
Type: Open Source

BentoML
✓ Model serving
✓ Package ML models
✓ Easy deployment
Type: Open Source

Ray Serve
✓ Scalable ML serving
✓ Distributed deployment
Type: Open Source

TorchServe
✓ PyTorch serving
✓ Production ready
Type: Open Source

vLLM
✓ High-throughput LLM serving
✓ Optimized inference
Type: Open Source

9. Container & Orchestration

Docker
✓ Containerization
✓ Consistency
Type: Open Source

Kubernetes
✓ Container orchestration
✓ Auto-scaling
Type: Open Source

Helm
✓ K8s package manager
✓ Deployment templates
Type: Open Source

PHASE 4: MONITORING & OPERATIONS

10. Experiment Tracking & Model Registry

MLflow
✓ Experiment tracking
✓ Model registry
✓ Widely used MLOps
Type: Open Source

Weights & Biases
✓ Experiment visualization
✓ Hyperparameter tracking
✓ Excellent dashboards
Type: Free + Enterprise

Neptune AI
✓ Detailed tracking
✓ Model debugging
Type: Paid

11. Monitoring & Observability

Langfuse
✓ LLM tracing
✓ Prompt monitoring
✓ LLM-specific observability
Type: Open Source + Cloud

Arize AI
✓ Model performance monitoring
✓ Drift detection
✓ ML observability
Type: Paid

WhyLabs
✓ Data drift monitoring
✓ Model health checks
✓ Data quality monitoring
Type: Paid

12. LLM Evaluation & Testing

DeepEval
✓ Automated LLM evaluation
✓ Benchmarking
Type: Open Source

Ragas
✓ RAG evaluation
✓ Answer faithfulness
Type: Open Source

OpenAI Evals
✓ LLM benchmarking
✓ Custom evaluations
Type: Open Source

CROSS-CUTTING CONCERNS

13. Guardrails & AI Safety

NeMo Guardrails
✓ Control LLM outputs
✓ Define safety rules
✓ Conversation rules
Type: Open Source

Guardrails AI
✓ Validate structured outputs
✓ Policy constraints
✓ Schema enforcement
Type: Open Source + Paid

Lakera Guard
✓ Prompt injection protection
✓ LLM security
✓ Threat detection
Type: Paid

LLM Guard
✓ Input/output scanning
✓ Content moderation
Type: Open Source

14. Workflow Orchestration & Pipelines

Kubeflow
✓ ML pipeline orchestration
✓ Automated retraining
✓ End-to-end ML workflows
Type: Open Source

Apache Airflow
✓ Schedule data pipelines
✓ Workflow automation
✓ Mature orchestration
Type: Open Source

Prefect
✓ Modern workflow engine
✓ Dynamic pipelines
Type: Open Source + Cloud

Dagster
✓ Data orchestration
✓ Asset-based pipelines
Type: Open Source + Cloud

15. CI/CD & Infrastructure as Code

GitHub Actions
✓ CI/CD workflows
✓ Automated testing
Type: Free + Enterprise

GitLab CI/CD
✓ Integrated pipelines
✓ Auto DevOps
Type: Open Source + Paid

Terraform
✓ Infrastructure as Code
✓ Multi-cloud provisioning
Type: Open Source + Enterprise

Ansible
✓ Configuration management
✓ Automation
Type: Open Source

SUPPORTING TOOLS

16. Cloud Providers & Compute

AWS
✓ SageMaker
✓ Bedrock
✓ EC2/ECS/EKS
Type: Cloud Service

Azure
✓ Azure ML
✓ OpenAI Service
✓ Cognitive Services
Type: Cloud Service

Google Cloud
✓ Vertex AI
✓ TPUs
✓ GKE
Type: Cloud Service

17. Testing & Code Quality

Pytest
✓ Python testing
✓ Fixtures
Type: Open Source

Black
✓ Code formatting
Type: Open Source

Ruff
✓ Fast linting
Type: Open Source

Pre-commit
✓ Git hooks
Type: Open Source

SECURITY & COMPLIANCE

18. Security & Secrets Management

HashiCorp Vault
✓ Secrets management
✓ Encryption as a service
✓ Access control
Type: Open Source + Enterprise

AWS Secrets Manager
✓ Rotate secrets
✓ AWS integration
Type: Cloud Service

Snyk
✓ Vulnerability scanning
✓ Dependency security
✓ Container scanning
Type: Free + Paid

19. Compliance & Governance

Model Cards
✓ Model documentation
✓ Transparency
Type: Framework

Fairlearn
✓ Bias detection
✓ Fairness metrics
Type: Open Source

SHAP
✓ Model explainability
✓ Feature importance
Type: Open Source

LEGEND & QUICK REFERENCE

Open Source - Free to use, community-driven

Paid/Cloud Service - Commercial platform or managed service

Hybrid - Open source core with enterprise/cloud paid options

✓ = *Key capability* ⚡ = *Performance optimized* 🔒 = *Security/Safety focused*